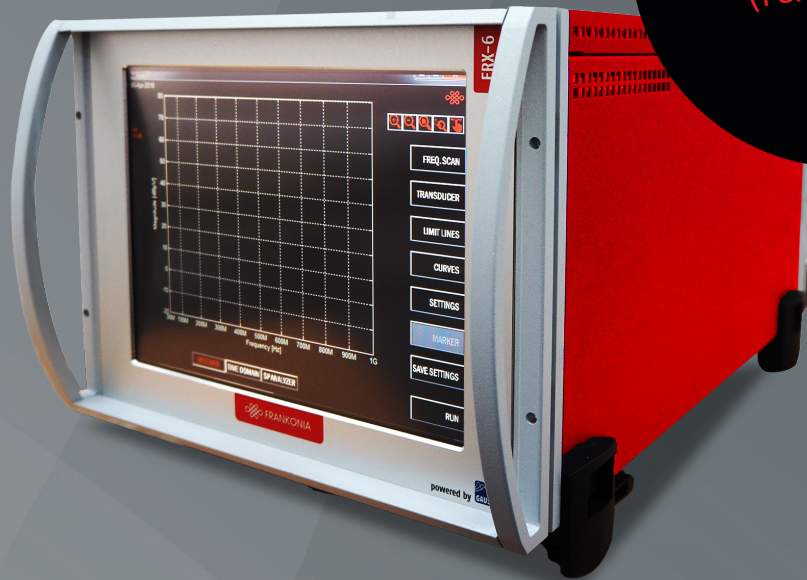




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EMC Test-Systems GmbH

ERX-6

DATA SHEET



10 Hz - 6 GHz
(7 GHz as Option)

Why FFT?

- ✓ EMI testing is time consuming
- ✓ Complex devices show a lot of different operation modes
- ✓ Introduction of FFT-based measuring instruments in CISPR 16-1-1 in 2010
- ✓ Introduction of new measurement strategies in CISPR 16-2-X in 2010
- ✓ FFT-based measuring instruments are also validly in further standards e.g. MIL-STD 461 and ETSI standards

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DESCRIPTION

The new Full-Compliance ERX-6 EMI test receiver from Frankonia is suitable for mobile use as well as for use in systems. The ERX-6 offers a very high measuring speed so that e.g. the measurement is reduced from hours to seconds with the Quasipeak detector for emission measurements.

By default, the ERX-6 is equipped with hardware that significantly accelerates the measurement according to the standards CISPR 16-1-1 Ed 3.1 and MIL461G (FFT-based measuring instrument). For standards that do not provide acceleration using the FFT-based measuring instrument, the ERX-6 has a classic stepped scan mode. In addition, it is possible to perform measurements at individual frequency points, as described e.g. at the final maximization.

In conjunction with the PROVE-EMC software, the receiver can significantly improve the measurement quality and reduce the test times in conjunction with the antenna mast and turntable. Due to the excellent noise floor of < -12 dBuV (band C / D) or -3 dBuV (band E) even difficult measuring tasks can be solved, without the need for an external preamplifier. The functional scope is rounded off by a very fast spectrum analyzer as well as a high-resolution scope function.



FRONT VIEW



REAR VIEW





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TECHN. SPECIFICATIONS	GENERAL DATA
Level	• Display range displayed noise floor up to +30 dBm • Maximum DC input level, pulse 6 V (0dB Att) • RF-CW signal 120 dBμV (13 dBm)
Intermodulation	• 1dB Compression Point of Mixer f < 1 GHz f > 1 GHz • Third order Intercept Point (TOI) 10 Hz – 6 GHz Typ. > 20dBm • Second Harmonic Intercept Point (SHI) 10 Hz – 6 GHz Typ. > 55dBm
Displayed Average Noise Level RF attenuation 0dB, normalized to 1 Hz RBW, mean marker (typ.)	• 9 kHz – 150 kHz: - 130 dBm • 150 kHz – 1 MHz: - 154 dBm • 1 MHz – 30 MHz: - 161 dBm • 30 MHz – 1000 MHz: - 163 dBm (LNA o) • 169 dBm (LNA on) • 1000 MHz – 6 GHz: -164 dmb (LNA o) • 170 dBm (LNA on)
Noise Indication RF attenuation 0dB, RBW according to CISPR 16-1-1, mean marker (typ.)	• 9 kHz – 150 kHz: < 0 dBuV • 150 kHz – 1 MHz: < -5 dBuV • 1 MHz – 30 MHz: < -18 dBuV • 30 MHz – 1000 MHz: < -6 dBuV (LNA o) < -12 dBuV (LNA on) • 1000 MHz – 6 GHz: < 4 dBuV (LNA o) < -3 dBuV (LNA on)
CISPR Indication Range	• 6 dB margin to noise oor over complete amplitude range according to CISPR 16-1-1 Ed. 3.1 • Quasi-peak indication according to CISPR 16-1-1: All pulse repetition frequencies • Peak, Average, CISPR-AVG indication according to CISPR 16-1-1 in all modes • CISPR-RMS indication according to CISPR 16-1-1 • Maximum deviation for sinusoidal signals according to CISPR 16-1-1: 2dB (9 kHz – 6 GHz)



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TECHN. SPECIFICATIONS	ERX-6
Frequency range	10 Hz – 6 GHz
Spectral purity	• SSB phase noise frequency = 500 MHz, carrier offset • 100 Hz < -100 dBc (1 Hz) • 1 kHz < -107 dBc (1 Hz) • 10 kHz < -101 dBc (1 Hz) • 100 kHz < -126 dBc (1 Hz) • 1 MHz < -146 dBc (1 Hz) • 10 MHz < -150 dBc (1 Hz) (nom.) • Residual FM frequency = 500 MHz, RBW = 1 kHz, Sweep time = 100 ms < 3 Hz (nom.)
Operating modes	• EMI receiver (superheterodyne) • EMI receiver (FFT-based measuring instrument) • Spectrum analyzer
Reference Oscillator	• Reference oscillator OCXO Aging < +/- 3.5 ppm / 15 years Temperature drift (0 – 60° C) < +/- 1 x 10e-8 SSB phase noise (1 Hz BW): 1 Hz -95 dBc/Hz (12.8 MHz) 10 Hz -120 dBc/Hz 100 Hz -140 dBc/Hz 1 kHz -145 dBc/Hz • Frequency resolution 0.01 Hz



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EMI-RECEIVER FFT-BASED MEASURING INSTRUMENT		ERX-6	
Frequency segment processed in parallel	• RBW = 10 Hz 0.06 MHz • RBW = 100 Hz 0.6 MHz • RBW = 200 Hz 1.1 MHz • RBW = 1 kHz 5.7 MHz • RBW = 9 kHz 57 MHz • RBW = 10 kHz 57 MHz • RBW = 120 kHz 225 MHz • RBW = 100 kHz 225 MHz • RBW = 1 MHz 225 MHz • RBW = 10 MHz 225MHz		
Scanning speed (Receiver mode) typ.	• Band A, Quasi-peak, dwell time 1 s : 2s • Band B, Quasi-peak, dwell time 1 s: 2 s • Band C/D Quasi-peak, dwell time 1 s: 30 s • Band E (1 GHz – 6 GHz), dwell time 100 ms: 2 s		
FFT-Overlapping Factor	• according to CISPR 16-1-1 and CISPR 16-3 • Overlapping factor typ > 95%		
FFT-BASED MEASURING INSTRUMENT ACCORDING TO CISPR 16-1-1, MIL461 AND OTHER EMC STANDARDS. SOMETIMES CALLED TIME-DOMAIN SCAN.			
Preselection and Preamplifier Structure	• Multiple paths with fixed filters		
Analog Preselection	• 10 Hz – 9 kHz • 9 kHz – 150 kHz • 150 kHz – 30 MHz • 30 MHz – 225 MHz • 225 MHz – 450 MHz • 450 MHz – 675 MHz • 675 MHz – 900 MHz • 900 MHz – 1 GHz • 1 GHz – 3 GHz • 3 GHz – 6 GHz		



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EMI-RECEIVER	ERX-6
Frequency readout (Analyzer mode)	• Marker resolution 0.5 Hz • Uncertainty $\pm(\text{marker frequency} \times \text{reference accuracy}$ $+ 10 \% \times \text{resolution bandwidth}$ $+ \frac{1}{2} (\text{span}/(\text{sweep points} - 1)) + 0.5 \text{ Hz}$ • Spectrum analyzer 1 to 8 000 000 (64 bit operation system) • EMI measurement 1 to 8 000 000 (64 bit operation system) • Marker tuning frequency step size marker step size = sweep points span/(sweep points - 1) • Marker step size = standard span/(default sweep points - 1) • Frequency counter resolution 0.001 Hz • Count accuracy $\pm (\text{frequency} \times \text{reference accuracy} + \frac{1}{2} (\text{last digit}))$ • Display range for frequency axis 0 Hz, 10 Hz to max. frequency • Resolution 0.1 Hz • Max. span deviation $\pm 0.1 \%$
Receiver scan	• Scan scan with max. 100 subranges with different settings • Scan modes normal scan, FFT-based measuring instrument according to CISPR 16-1-1 • Measurement time superhet scan, per frequency 1 s to >100 s • Measurement time FFT-based measuring instrument, per frequency 1 s to >100 s • Number of trace points up to 8 000 000 • Frequency step size normal scan min. 1 Hz • Frequency



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Preamplifier	• Located between preselection and ADC & mixer • Individual pre-amps for different frequency ranges • Gain: Typ. 20 dB, full overrange detection and auto attenuation • Noise figure: Typ 3.5 dB • Frequency range: 10 Hz – 6 GHz
Digital Preselection	• 0 MHz – 225 MHz • 225 MHz – 450 MHz • 450 MHz – 675 MHz • 675 MHz – 900 MHz • 900 MHz – 1 GHz
Low Noise Amplifier	• 30MHz – 6 GHz (7 GHz with Option ERX-FE7), Noise Figure < 4dB

SPECTRUM ANALYZER

Spectrum Analyzer	• Sweep time range span = 0 Hz, 1 s to 16000 s • Span 10 Hz, swept 1 us to 16000 s • Span 10 Hz, FFT based measuring instrument 1 s to 16000 s • Sweep time accuracy span = 0 Hz ± 0.1 % (nom.) • Span 10 Hz, swept ± 1 % (nom.)
IF Bandwidths	• 3dB bandwidth: 1 Hz – 30 MHz • 1, 2, 3, 5 steps • Small step size (145 steps) for channel measurements • 6dB bandwidths CISPR: 200 Hz, 9 kHz, 120 kHz, 1 MHz • 6dB bandwidths MIL/DO: 10 Hz, 100 Hz, 1 kHz, 100 kHz, 1 MHz
Videofilter	• 0.1 Hz – 10 MHz + 3*RBW • 1, 2, 3, 5 steps • Detectors: MaxPeak, MinPeak, Sample
Detectors (Videofilter off / 3 RBW)	• Maxpeak, Average, RMS • Dynamic requirements according to CISPR 16-1-1 (Peak, AVG)



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TECHN. SPECIFICATIONS	GENERAL DATA
Absolute level uncertainty	• Signal level : 40 – 60 dBuV (15 MHz) < 0.3 dB (= 0.1) • Frequency response (9 kHz – 6 GHz) < 0.5 dB (= 0.15) • Attenuator switching uncertainty (15 MHz) < 0.2 dB (= 0.15) • Total measurement uncertainty CW signal, S/N > 20 dB 95 % confidence level: 0.5 dB
Spurious Response	• Residual spurious response RF attenuation = 0 dB • $f \leq 1$ MHz < –107 dBm • $f \leq 1$ MHz < –117 dBm • $f > 1$ MHz < –112 dBm • Image frequency < –80 dBc (nom.)
Attenuator	• 0 – 50 dB • 10 dB steps
Input Ports (RF Input)	• N type connector, 50 Ohm • 0 dB attenuator: VSWR < 2.0 (9 kHz – 1 GHz) • 10 dB attenuator: VSWR < 3.0 (1 GHz – 6 GHz)
Remote Control	Remote control command set according to SCPI standard
PC	• Mobile Multi Core processor, 8 GByte RAM, • >250 GByte Solid State Disc • Operation system: Windows® 10, 64Bit
Power Supply	• +11 V .. +14 V DC, 230 V +/- 20 % 50 Hz or 110 V +/- 10% 60 Hz • Max. power consumption approx. 60 W
Interfaces	• Ethernet/LAN, USB, VGA, HDMI, Audio
Display, User Interface	• Resolution 800 x 600 Pixel, 8,4" , TrueColor (16.78 Mio. colors), Touchscreen
Weight	• approx. 8 kg



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